

MonthlyGrocery – AI Architecture & Operational Guide

Comprehensive Specification for Multimodal Ingestion, Conversational Planning & Predictive Commerce

1. Strategic Positioning: India's Digital Monthly Grocery Assistant

Unlike 10-minute quick-commerce platforms designed for impulse daily top-ups, MonthlyGrocery solves the **planned recurring monthly household basket** (₹3,000–₹10,000 AOV). Indian households buy staples in bulk once a month. The AI engine acts as a personal grocery manager: digitizing paper slips, predicting refill cycles, optimizing bundle savings, and building baskets with zero manual effort.

2. Three-Tier Architectural Integration with AI Layer

Super Admin Layer

Master SKU catalog, global pricing rules, AI taxonomy and embedding generation.

- Auto-categorization & HSN code detection
- Master product vector embeddings
- Global brand substitution mappings

Area Admin Layer

Local inventory, area-level discounts, delivery slots, localized SKU filtering.

- Area-specific stock & price filtering
- Demand forecasting for 1st–5th restock
- Local deal & coupon injection

Customer Mobile / Web

End-user interfaces for multi-modal ingestion, conversational planning & 1-click cart.

- OCR slip scanner & voice shopping
- Interactive conversational assistant
- Personalized basket & savings alerts

3. End-to-End AI Data Flow & Pipeline Topology

[Customer Inputs: Handwritten Slip / Hinglish Audio / WhatsApp Text / Chat Prompt]

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(Express Backend API: /api/ai/...)

AI Perception & Parsing Layer (Google Gemini 1.5 Flash / Whisper Speech API) |
• Vision OCR: Extract handwritten line items + brand + weight + pack count |
• Audio NLP: Transcribe Hinglish audio & parse colloquial entity intents |
• Structured JSON normalizer: [{ query_item: "Atta", qty: 10, unit: "kg" }, ...] |

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Semantic Matching & SKU Resolution Engine |
• Filter SKUs by Customer's active `area_id` & in-stock status |
• Vector Embedding Match (pgvector / cosine similarity) + Trigram fuzzy text matching |
• Brand Affinity Weighting (Dove vs Santoor preference based on order history) |

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Smart Savings & Pack Arbitrage Optimizer |
• Compare unit economics (₹/kg on 1kg vs 5kg/10kg bulk packs) |
• Inject active Area Admin promotions ("Buy ₹2500+ unlock free 1kg sugar") |

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[Live Verified Cart with High-Confidence Auto-Adds & Reviewable Matches in UI]

4. Summary of Planned AI Modules

Module Name	Type	Core User Action	Underlying Technology
1. Handwritten OCR Scanner	PERCEPTION	Upload photo of physical paper grocery list	Gemini 1.5 Flash Vision + Semantic Catalog Matcher
2. Hinglish Voice Shopping	SPEECH/NLP	Speak voice note: "10kg atta, 5L oil, 2kg sugar"	Whisper API / Google Cloud Speech + Intent Parser
3. WhatsApp Text List Parser	NLP	Paste raw copied text notes into the app	Few-shot entity parsing model with unit normalizer
4. Conversational Assistant	CHAT/LLM	Chat: "Plan 4-person veg monthly grocery under ₹5k"	Gemini 1.5 with Function Calling & Catalog Grounding
5. Household Refill Predictor	PREDICTIVE	Auto-receive "Your monthly grocery is due" alert	Inter-purchase consumption velocity heuristics
6. Pack-Size Arbitrage Engine	OPTIMIZATION	Display "Save ₹110 by switching to 5kg pack"	Deterministic unit price calculator + dynamic upsell

Multi-Modal "Grocery List-to-Cart" Ingestion Engine

Handwritten Slip OCR, Vernacular Voice Notes & Raw Text Entity Resolution

Module Code: MG-AI-01
Target UI: Mobile App & Web

1. Handwritten Paper Grocery Slip Scanner (OCR + Vision LLM)

Workflow Steps

- Image Capture:**
User photographs their paper list (camera/gallery). Compressed to <2MB on device.
- Vision Entity Extraction:**
Image sent to POST `/api/ai/ocr-list`. Gemini 1.5 Flash Vision extracts handwriting in structured JSON.
- Area SKU Matching:**
Normalizer maps extracted text against Area Admin's active stock using trigram + embeddings.
- Cart Review Screen:**
Products matched with >85% confidence are auto-selected; ambiguous items show 2-3 candidate chips.

Sample Vision Extraction Prompt & JSON Output

```
// System Prompt: Extract grocery items, brands, and quantities
{
  "raw_items_detected": [
    { "text": "Aashirwad atta 10kg", "name": "Atta", "brand": "Aashirvaad", "qty": 10, "unit": "kg" },
    { "text": "Fortune oil 5 ltr", "name": "Sunflower Oil", "brand": "Fortune", "qty": 5, "unit": "l" },
    { "text": "Sugar 5kg", "name": "Sugar", "brand": null, "qty": 5, "unit": "kg" },
    { "text": "Surf excel 2kg", "name": "Detergent", "brand": "Surf Excel", "qty": 2, "unit": "kg" }
  ]
}
```

2. Voice Shopping & Hinglish Audio-to-Cart

Speech Pipeline Flow

Enables natural Hindi / Hinglish / English voice notes (e.g., "Bhaiya 10 kilo rice, 5 kilo atta, 2 packet Tata Namak aur 1 kilo Moong Dal daal do").

- Audio Recording:** High-quality AAC/WAV audio stream recorded via mobile mic (max 30s).
- STT Transcription:** Transcribed via Whisper API or Google Speech API with Indian accent acoustics.
- Colloquial Normalization:** Maps terms like "dabba", "packet", "theli", "kilo", "pao" to standardized metric units.

Colloquial Entity Mapping Table

Spoken Term	Normalized Unit	Default Pack Mapping
"1 pao" / "ek pao"	250 grams	250g SKU
"adha kilo"	500 grams	500g SKU
"ek theli / 1 packet doodh"	500 ml / 1L	Standard Pouch SKU
"chhota dabba"	Small / 200g	Lowest SKU weight

3. SKU Resolution & Ambiguity Management Algorithm

Multi-Stage Ranking Pipeline

When an item like "Sugar 5kg" is requested without a brand, the system resolves it using this mathematical priority:

Step A: User Purchase History

If user previously bought *Madhur Pure & Hygienic Sugar 5kg* in last 90 days, select Madhur (\$w = 0.50\$).

Step B: Area Admin Best Seller

If new user, rank by highest sales volume in user's active `area_id` (\$w = 0.30\$).

Step C: Max Margin / Value Deal

Tiebreaker prioritizes active bundle offer or higher savings margin (\$w = 0.20\$).

Conversational AI Grocery Assistant & Budget Solver

Natural Language Household Planning, Budget Constrained Solvers & Value Optimizers

Module Code: MG-AI-02
Target UI: In-App Assistant & Web Chat

1. Conversational Engine Architecture & Guardrails

System Prompt Guidelines

The assistant acts as an expert Indian grocery planning consultant. It strictly grounds responses in available products from the user's selected Area Admin.

- Grounding:** Only recommend products present in current area's active stock.
- Language:** Seamlessly respond in English, Hindi, or Hinglish matching user tone.
- Actionable Tool Execution:** Instead of long text lists, directly call backend function `generate_monthly_basket()` to render live interactive cart cards.

Function Calling / Tool Schema

```
{
  "name": "generate_monthly_basket",
  "description": "Constructs or modifies user monthly grocery cart",
  "parameters": {
    "type": "object",
    "properties": {
      "target_budget": { "type": "number" },
      "dietary_preference": { "type": "string", "enum": ["veg", "non-veg", "jain"] },
      "items": {
        "type": "array",
        "items": {
          "type": "object",
          "properties": {
            "sku_id": { "type": "string" },
            "quantity": { "type": "integer" },
            "reason": { "type": "string" }
          }
        }
      }
    }
  }
}
```

2. Three Core Conversational Scenarios & Solver Logic

Scenario A: Family Profile Grocery Generator

User Request: "Humare ghar mein 4 log hain (2 adults, 2 kids, veg). 1 month ki grocery bana do."

AI Action: Invokes consumption formula for 4-member veg family: 15kg Atta, 10kg Rice, 5L Oil, 4kg Dal assortment, 5kg Sugar, cleaning essentials. Generates interactive cart for ~₹4,500.

Scenario B: Hard Budget-Constrained Basket Solver

User Request: "Mujhe strict ₹5,000 ke budget ke andar poori monthly grocery set karni hai."

AI Action: Runs linear knapsack optimization allocating budget across primary categories (Staples: 55%, Oils: 20%, Cleaning: 15%, Snacks: 10%). Maximizes caloric and essential utility while capping total cost ≤ ₹5,000\$.

Scenario C: Budget Reduction & Savings Maximizer

User Request: "Mere last month ke order se ₹500 kam mein poori basket bana do."

AI Action: Fetches last month's basket (₹4,860). Performs 3-step value arbitrage: (1) Converts 1kg packs to 5kg bulk packs (-₹140), (2) Replaces branded premium pulses with certified value grade (-₹180), (3) Applies active coupon code (-₹180). New total: ₹4,360 (Saved ₹500!).

3. Interactive UI Render Component

AI-Generated Basket Preview Card (In-Chat UI)

CALCULATED SAVINGS: ₹465

Staples (Atta/Rice)
₹1,850 (25 kg)Edible Oils & Ghee
₹920 (5 Litres)Pulses & Spices
₹840 (8 Items)Home & Cleaning
₹740 (5 Items)

Total Basket Value: ₹4,350 ₹4,815 MRP

[Transfer All to Cart]

Household Consumption Modeling & Predictive Refills

Algorithmic Household Baseline Calculations, Consumption Velocity & Automated Reminders

Module Code: MG-AI-03
Target UI: Push Notifications, WhatsApp & Profile Hub

1. Household Algorithmic Consumption Engine

During customer onboarding, the app collects demographic parameters: **Adults (\$A\$)**, **Children (\$C\$)**, **Seniors (\$S\$)**, **Diet Preference (\$D\$)**, and **Cooking Frequency (\$F\$)**. The consumption engine computes baseline quantity $Q_{category}$ using empirical household standards:

Category	Consumption Formula (per month)	Example (2A + 2C)
Atta / Wheat Flour	$Q = (A \times 3.5) + (C \times 2.0) + (S \times 2.5) \text{ ext\{kg\}}$	11.0 kg (10kg + 1kg pack)
Rice	$Q = (A \times 2.5) + (C \times 1.5) + (S \times 2.0) \text{ ext\{kg\}}$	8.0 kg (5kg + 3kg packs)
Cooking Oil	$Q = (A \times 1.2) + (C \times 0.6) + (S \times 0.8) \text{ ext\{Litres\}}$	3.6 Litres (1 \$imes\$ 5L Jar)
Sugar & Salt	$Q_{sugar} = (A+C) \times 1.0 \text{ ext\{kg\}}$; $Q_{salt} = 1 \text{ ext\{kg fixed\}}$	4.0 kg Sugar, 1 kg Salt
Dal & Pulses	$Q = (A \times 1.2) + (C \times 0.6) + (S \times 1.0) \text{ ext\{kg\}}$	3.6 kg (Toor, Moong, Chana)
Detergent / Laundry	$Q = 1.0 \text{ ext\{kg\}} + (C \times 0.75 \text{ ext\{kg\}})$	2.5 kg (2kg + 500g)

Dietary & Regional Adjustments

- South India Region:** Rice weight \$+40\%\$, Atta weight \$-50\%\$, Mustard oil replaced with Sunflower/Gingelly.
- North India Region:** Atta weight \$+30\%\$, Rice weight \$-25\%\$, Mustard / Ghee preference boosted.
- Jain Preference:** Automatically excludes root vegetable derivatives, garlic-containing seasonings, and gelatin items.

2. Predictive Refill Intelligence & Replenishment Velocity

Velocity Tracking Algorithm

For each customer and recurring SKU category, the engine calculates the dynamic replenishment cycle:

```
Interval  $T_{cycle} = \frac{\sum_{i=1}^n (t_i - t_{i-1})}{n}$   
Days Elapsed  $D_{elapsed} = \text{ext\{CurrentDate\}} - t_{last\_order}$   
Refill Trigger when  $D_{elapsed} \geq (T_{cycle} - 3 \text{ ext\{days\}})$ 
```

If a customer buys 10kg Atta every 28 days, on Day 25 the proactive refill workflow triggers automatically.

Automated Alert Templates

WhatsApp & Push: "Monthly Grocery Due"

"Namaste Sharma ji! 🙏 Aapka monthly grocery due ho gaya hai. Aapke pichle 3 months ke hisab se 24 items ki basket ready hai. **Save ₹425** with today's offers. [Click to Review Cart]"

Micro-Nudge: "Running Low on Essentials"

"You usually run out of Tata Salt around Day 30. Would you like to add 1kg to your pending cart?"

3. Retention & Repeat Ordering Metrics

KPI Metric	Target Baseline	AI Optimization Impact
30-Day Repeat Purchase Rate	25% - 30%	45% - 55% via automated refill nudges
Cart Abandonment Reduction	65% average	<35% with pre-filled 1-click cart
Average Order Value (AOV)	₹1,800 (ad-hoc)	₹4,500 - ₹5,500 (complete household bundle)

One-Click Monthly Cart, Savings Arbitrage & Admin AI

Brand Affinity Matrices, Pack Size Economics & Area Admin Demand Forecasting

Module Code: MG-AI-04
Target UI: Home Screen CTA, Admin Dashboard

1. One-Click Monthly Cart & User Brand Affinity Matrix

Hero Feature: "Create My Monthly Cart"

When the user taps the prominent "MY MONTHLY GROCERY" CTA, the backend synthesizes their ideal basket in <500ms using a brand affinity score:

$$SP(\text{Brand}_i \mid \text{Category}_k) = \frac{\text{Orders with Brand}_i \cap \text{Category}_k}{\text{Total Category Orders}}$$

If a customer consistently orders Dove soap, the 1-Click Cart selects Dove; if another buys Santoor, it selects Santoor.

Out-of-Stock Smart Replacement Logic

If the customer's preferred brand is out-of-stock in their Area Admin's store, the AI does not leave a blank:

- Selects closest match by brand tier & price point.
- Renders a visible badge in UI: "Substituted Aashirvaad with Fortune Atta due to stock unavailability [Change]".

2. Smart Savings & Pack-Size Arbitrage Engine

Bulk Pack Unit Economics

Indian grocery shoppers are value-conscious. The engine dynamically detects inefficient cart compositions and displays an instant upgrade badge:

Customer Selection	AI Arbitrage Suggestion	Instant Saving
5 \$imes\$ 1kg Atta (₹250)	1 \$imes\$ 5kg Pack (₹215)	Save ₹35
4 \$imes\$ 500ml Oil (₹480)	1 \$imes\$ 2L Jar (₹410)	Save ₹70
5 \$imes\$ 1kg Sugar (₹240)	1 \$imes\$ 5kg Bag (₹195)	Save ₹45

Savings Zone & Promotion Engine

Connects cart total with Super Admin / Area Admin promotional campaigns:

- Threshold Unlock:** "Add ₹340 more to unlock Free 1kg Sugar & Free Delivery!"
- MonthlyCoins Cashback:** Automatically calculates earned loyalty coins (e.g. ₹5,000 order = 100 MonthlyCoins).

3. Admin & Operational AI Capabilities

Super Admin: Smart SKU Onboarding

When Super Admin uploads raw product images or Excel sheets, AI automatically generates:

- Standardized title & SEO metadata
- Category & subcategory tagging (15-20 core categories)
- Nutritional facts & dietary flags (Veg, Vegan, Gluten-Free)

Area Admin: Restock Demand Forecasting

90% of monthly grocery orders arrive between the 1st and 5th of each month (salary cycle).

- Forecasts total required volume 5 days in advance (Day 25).
- Prevents out-of-stock incidents for top 50 staple SKUs.

API Contracts, PostgreSQL Schemas & Implementation Plan

Backend Route Definitions, Relational Data Models & Phase-wise Milestone Schedule

Target Backend: Express / Node.js
Database: PostgreSQL + pgvector

1. Dedicated AI RESTful API Endpoints

Endpoint	Method	Request Payload Summary	Response Summary
/api/ai/ocr-list	POST	multipart/form-data (image file, area_id)	Array of extracted items mapped to active SKUs with confidence scores
/api/ai/voice-to-cart	POST	multipart/form-data (audio file, area_id)	Parsed entities, normalized quantities, resolved cart items
/api/ai/assistant/chat	POST	{ user_id, message, conversation_id, area_id }	Assistant response + structured action cards (basket payload)
/api/ai/monthly-basket	POST	{ user_id, area_id, budget_limit, family_id }	Optimized 1-Click suggested monthly basket with savings breakdown
/api/ai/predictive-refill	GET	Query param: user_id	List of items due for replenishment with cycle indicators

2. Database Schema Extensions (PostgreSQL)

User Household Profiles Table

```
CREATE TABLE user_household_profiles (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  user_id UUID REFERENCES users(id) ON DELETE CASCADE,  
  adults_count INT DEFAULT 2,  
  children_count INT DEFAULT 0,  
  seniors_count INT DEFAULT 0,  
  dietary_preference VARCHAR(30) DEFAULT 'veg', -- veg/non-veg/jain  
  monthly_budget NUMERIC(10,2),  
  preferred_brands JSONB DEFAULT '[]'::jsonb,  
  created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()  
);
```

User Consumption Cycles Table

```
CREATE TABLE user_consumption_cycles (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  user_id UUID REFERENCES users(id) ON DELETE CASCADE,  
  category_id UUID REFERENCES categories(id),  
  sku_id UUID REFERENCES master_products(id),  
  avg_cycle_days INT NOT NULL DEFAULT 30,  
  last_purchase_date TIMESTAMP WITH TIME ZONE,  
  predicted_next_refill TIMESTAMP WITH TIME ZONE,  
  last_notification_sent TIMESTAMP WITH TIME ZONE  
);
```

3. Implementation Milestones & Delivery Roadmap

Phase & Timeline	Milestone Deliverables	Primary Tech Components
Phase 1 (Sprint 1-2) Quick Ingestion	- Handwritten Slip OCR Scanner & UI Review Screen - Raw WhatsApp / Text Notes List-to-Cart Parser - Pack-Size Savings Arbitrage suggestions in Cart	Gemini 1.5 Flash Vision, Express OCR routes, Mobile camera UI
Phase 2 (Sprint 3-4) Conversational & Profile	- Hinglish Voice Shopping (Audio Note to Cart) - Conversational Grocery Assistant (In-App Chat) - Household Demographics Profile & Consumption Calculator	Whisper STT / Speech API, Gemini Function Calling, Chat Screen
Phase 3 (Sprint 5-6) Predictive Retention	- Automated Refill Prediction & WhatsApp Reminders - One-Click Monthly Basket Generator with Brand Affinity - Area Admin Restock Demand Forecasting	Consumption Velocity Cron Jobs, WhatsApp Cloud API, pgvector